

Iron Sentinel: Full Attack Timeline

Simulated Intrusion - Reconstructed Timeline

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Overview

This document reconstructs the simulated attack carried out against the Iron Sentinel lab environment. Three MITRE ATT&CK techniques were used against the Windows Server 2022 host and captured by Sysmon and Windows Security Auditing. Splunk ingested all activity through the Universal Forwarder, and each technique was checked against its corresponding detection rule.

The summary tables below show what was simulated and how it was confirmed. Full detail on each technique follows.

	Technique Simulated	MITRE ID	Status
1	Brute Force (repeated failed logons)	T1110	Validated
2	Powershell Obfuscation	T1059.001	Validated
3	Off-Hour Login (Valid Account)	T1078	Logic Verified

Technique Detail

1. Brute Force - Repeated Failed Logons

MITRE ATT&CK: T1110 (Credential Access)

Six consecutive failed NTLM authentication attempts were generated against the local Windows Server 2022 host using a non-existent account.

Command Used:

```
for /L %i in (1,1,6) do net use \\localhost\C$ /user:baduser  
wrongpassword123
```

Evidence:

- Windows Security Event ID 4625 generated for each attempt- Failure Reason: "Unknown user name or bad password"
- Account_Name field consistently logged baduser
- 84 total Event ID 4625 events captured across the test window

Detection Result:

The Failed Login Brute Force Detection rule was run against the captured data using a 5-minute bucket window:

- index=main EventCode=4625 Account_Name=baduser |
bucket_time span=5m | stats count by _time, Account_Name |
where count > 5

2. Powershell Obfuscation

MITRE ATT&CK: T1059.001 (Execution)

A Base64-encoded PowerShell command was executed with flag commonly used to evade detection and hide the operator's intent.

Command Used:

- powershell.exe -nop -w hidden -enc
JABhACAAJwBoAGUAbABsAG8AJwA7AA==
 - Decodes to the harmless statement: \$a = 'hello' - used only to demonstrate the obfuscation pattern, not to deliver a payload)

Evidence:

Detection Result:

The PowerShell Obfuscation Detection Rule searches for the combination of obfuscation-related flags in the command line:

- index=main sourcetype="XmlWinEventLog:Sysmon"
" <EventID>1</EventID>" "powershell.exe" "-enc"

The simulated command appeared in the result set with its full command line intact, confirming the rule's pattern-matching logic correctly flags this obfuscation technique.

3. Valid Accounts - Off-Hour Login

MITRE ATT&CK: T1078 (Persistence/ Initial Access)

This technique watches for successful logons (Event ID 4624) that occur outside normal business hours, defined here as before 6 AM or after 8 PM

Detection Rule:

- index=main sourcetype="WinEventLog:Security" EventCode=4624
(date_hour<6 OR date_hour>20)

Validation Approach

A live off-hours trigger was attempted by adjusting the Windows VM system clock, but clock synchronization conflicts with the host machine prevented a clean, verifiable timestamp on the resulting event. Rather than rely on an unreliable test, the rule's logic and field extraction were validated directly against the lab's full logon history instead.

Evidence

- 720 total Event ID 4624 events reviewed across the lab's operating window
- All legitimate logons fell between 6 AM and 6 PM - all inside normal business hours
- The correctly returned zero results against this dataset, confirming no false positives during normal operation
- Date_hour field confirmed present and correctly extracted on all 4624 events

Rule logic is sound and field extraction is confirmed working. The rule will trigger correctly on any future logon outside the 6 AM-8 PM window. This is documented as logic-verified rather than live-fired, which is a standard and defensible validation method for time-boundary detections that are impractical to trigger safely in a lab.

Lessons Learned

- The Splunk Universal Forwarder went silent for several days (June 11-20) without an obvious alert. Forwarder health should be checked at the start of every session/day, and not assumed.
- Raw XML-formatted Sysmon data does not behave like extracted Splunk fields. Searches need to use raw text matching (ex. "powershell.exe") rather than field syntax (ex. Image=) unless field extraction is explicitly configured.
- Not every technique needs a live-fired trigger to be validated. Confirming rule logic against historical data is a legitimate method when live testing is impractical or unsafe.
- Third-party offensive security tooling (e.g. Atomic Red Team) is frequently flagged by antivirus as malicious by design. Manually scripting the equivalent test commands avoided this entirely and kept the host machine clean.